

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

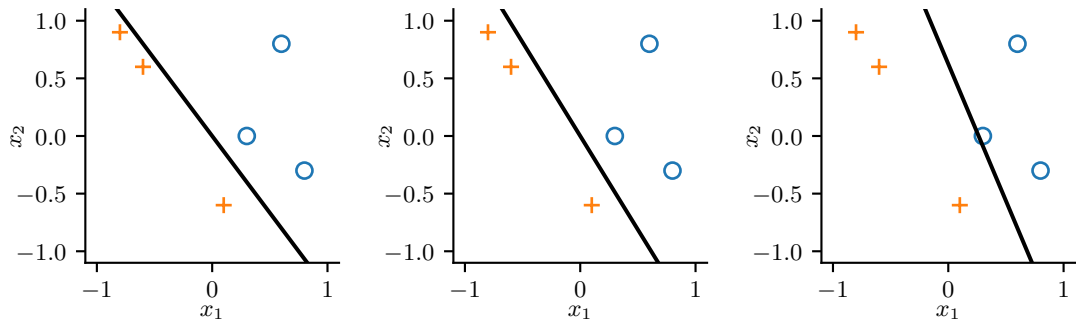
Slide Set 6: Logistic Regression



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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 - ▶ For use only by enrolled students in the course

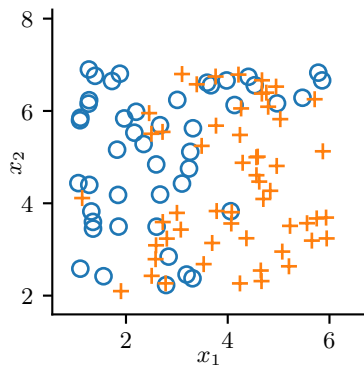
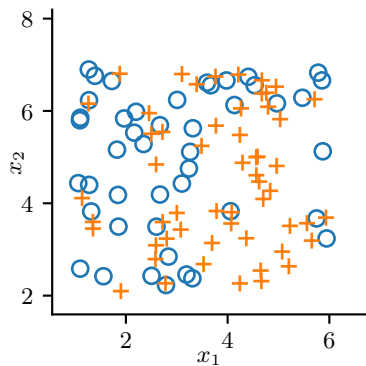
Calibrated Output

Consider these decision surfaces, each with zero training error.

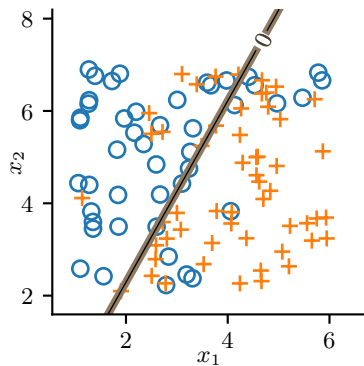
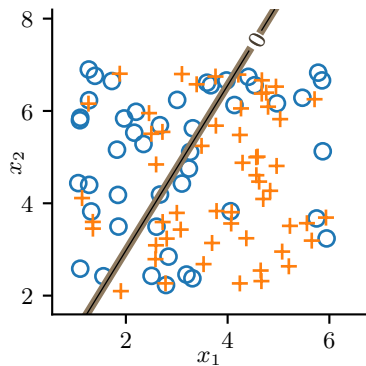


The perceptron learning algorithm makes no distinction among them.

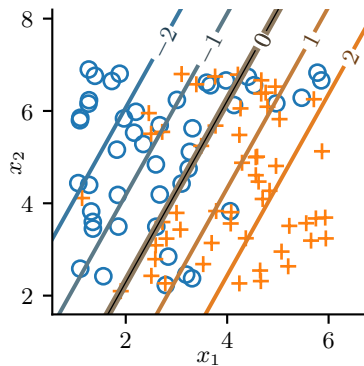
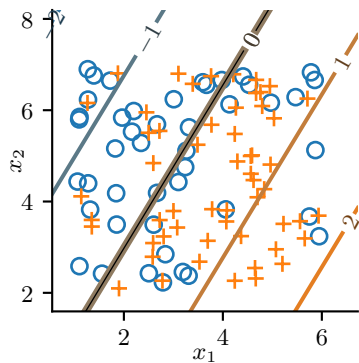
Calibrated Output



Calibrated Output



Calibrated Output



Sigmoid / Logistic

To make $w^\top x$ relate to the chance that the label is positive, we have to remap it from $(-\infty, \infty)$ to $(0, 1)$.

We use the “sigmoid” or “logistic” function:

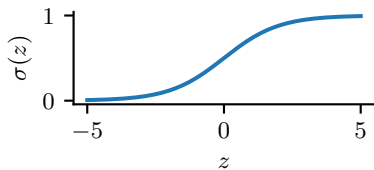
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid / Logistic

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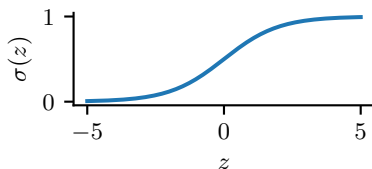


Sigmoid / Logistic

To make $w^\top x$ relate to the chance that the label is positive, we have to remap it from $(-\infty, \infty)$ to $(0, 1)$.

We use the “sigmoid” or “logistic” function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



Why?

If z is the log-odds of an event, then $\sigma(z)$ is the probability of the event.

Sigmoid Derivative

We will (later) need the derivative of $\sigma(z)$, so we'll do it now:

$$\begin{aligned}\sigma(z) &= \frac{1}{1 + e^{-z}} \\ \sigma'(z) &= \frac{-1}{(1 + e^{-z})^2} (-e^{-z}) \\ &= \frac{1}{1 + e^{-z}} \frac{e^{-z}}{1 + e^{-z}} \\ &= \frac{1}{1 + e^{-z}} \left(1 - \frac{1}{1 + e^{-z}} \right) \\ &= \sigma(z) (1 - \sigma(z))\end{aligned}$$

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Like saying that

$$g(z) = e^{2z} \quad \Rightarrow \quad g'(z) = 2g(z)$$

(Binary) Logistic Regression

Let $\tilde{f}(\vec{x}) = \vec{w}^\top \vec{x}$ be the discriminant.

(Binary) Logistic Regression

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$\tilde{f}(\vec{x})$ is the log-odds the class is positive.

Thus, $\sigma(\tilde{f}(\vec{x}))$ is the (predicted) probability the class is positive:

$$p(y = +1 \mid \vec{x}) = \frac{1}{1 + e^{-\vec{w}^\top \vec{x}}} = \sigma(\tilde{f}(\vec{x}))$$

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$$p(y = +1 \mid \vec{x}) = \frac{1}{1 + e^{-\vec{w}^\top \vec{x}}} = \sigma(\tilde{f}(\vec{x}))$$

Then the probability the class is negative is

$$p(y = -1 \mid \vec{x}) = 1 - \sigma(\tilde{f}(\vec{x})) = \frac{e^{-\vec{w}^\top \vec{x}}}{1 + e^{-\vec{w}^\top \vec{x}}} = \frac{1}{1 + e^{\vec{w}^\top \vec{x}}} = \sigma(-\tilde{f}(\vec{x}))$$

(Binary) Logistic Regression

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So, in general

$$p(y = y \mid \vec{x}) = \sigma(y\tilde{f}(\vec{x}))$$

(Binary) Logistic Regression

Goal: pick $\vec{w}$ so that the estimated probabilities make training y s most likely, given the $\vec{x}$ s

$$\arg \max_{\vec{w}} \prod_{i=1}^m p(y_i \mid \vec{x}_i)$$

(Binary) Logistic Regression

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$$\arg \max_{\vec{w}} \prod_{i=1}^m p(y_i \mid \vec{x}_i)$$

Same as

$$\arg \max_{\vec{w}} \sum_{i=1}^m \ln p(y_i \mid \vec{x}_i)$$

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Same as

$$\arg \min_{\vec{w}} - \sum_{i=1}^m \ln p(y_i \mid \vec{x}_i)$$

So,

$$L = \sum_{i=1}^m -\ln p(y_i \mid \vec{x}_i) = \sum_{i=1}^m -\ln \sigma(y_i \tilde{f}(\vec{x}_i))$$

(Binary) Logistic Regression

How to minimize

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

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$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

The gradient of L :

$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i)$$

$$L = \sum_{i=1}^m -\ln p_i$$

(Binary) Logistic Regression

How to minimize

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

The gradient of L :

$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i)$$
$$\frac{\partial L}{\partial w_k} = \sum_{i=1}^m \frac{-1}{p_i} \frac{\partial p_i}{\partial w_k}$$
$$L = \sum_{i=1}^m -\ln p_i$$

(Binary) Logistic Regression

How to minimize

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

The gradient of L :

$$\begin{aligned} p_i &= \sigma(y_i \vec{w}^\top \vec{x}_i) \\ L &= \sum_{i=1}^m -\ln p_i \end{aligned} \qquad \begin{aligned} \frac{\partial L}{\partial w_k} &= \sum_{i=1}^m \frac{-1}{p_i} \frac{\partial p_i}{\partial w_k} \\ &= \sum_{i=1}^m \frac{-1}{p_i} p_i (1 - p_i) \frac{\partial (y_i \vec{w}^\top \vec{x}_i)}{\partial w_k} \end{aligned}$$

(Binary) Logistic Regression

How to minimize

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

The gradient of L :

$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i)$$

$$L = \sum_{i=1}^m -\ln p_i$$

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(Binary) Logistic Regression

How to minimize

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

The gradient of L :

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(Binary) Logistic Regression Algorithm

Recall:

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

$$\nabla_{\vec{w}} L = \sum_{i=1}^m -(1 - p_i) y_i \vec{x}_i$$

$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i)$$

(Binary) Logistic Regression Algorithm

Recall:

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^T \vec{x}_i)$$

$$\nabla_{\vec{w}} L = \sum_{i=1}^m -(1 - p_i) y_i \vec{x}_i$$

$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i)$$

Gradient descent algorithm:

- ➊ Let $\vec{w}$ be a random weight vector
- ➋ While $\vec{w}$ is not at a local minimum of L
 - ➊ Let $\vec{g} \leftarrow 0$
 - ➋ For $i = 1, \dots, m$
 - ➊ Let $p_i \leftarrow \sigma(y_i \vec{w}^T \vec{x}_i)$
 - ➋ Let $\vec{g}_i \leftarrow -(1 - p_i) y_i \vec{x}_i$
 - ➌ Let $\vec{g} \leftarrow \vec{g} + \vec{g}_i$
 - ➌ Let $\vec{w} \leftarrow \vec{w} - \eta \vec{g}$

(Binary) Logistic Regression Algorithm

Recall:

$$L = \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i)$$

$$\nabla_{\vec{w}} L = \sum_{i=1}^m -(1 - p_i) y_i \vec{x}_i$$

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Stochastic gradient descent algorithm:

- ➊ Let $\vec{w}$ be a random weight vector
- ➋ While $\vec{w}$ is not at a local minimum of L
 - ➊ For $i = 1, \dots, m$
 - ➊ Let $p_i \leftarrow \sigma(y_i \vec{w}^\top \vec{x}_i)$
 - ➋ Let $\vec{g}_i \leftarrow -(1 - p_i) y_i \vec{x}_i$
 - ➌ Let $\vec{w} \leftarrow \vec{w} - \eta \vec{g}_i$

(Binary) Logistic Regression Optimization

Optimization notes:

- Unique global minimum, except
 - ▶ if data are separable, or
 - ▶ Features are linearly dependent
- More advanced optimization possible and often used
 - ▶ Second-order methods (uses second derivatives)
 - ▶ Does not require picking step sizes
 - ▶ Based on Newton's method
 - ▶ In this case, it is called iteratively reweighted least squares (IRLS)

Binary Classification Loss

Logistic regression loss is

$$\begin{aligned} L &= \sum_{i=1}^m -\ln \sigma(y_i \vec{w}^\top \vec{x}_i) \\ &= \sum_{i=1}^m -\ln \sigma(y_i \tilde{f}(\vec{x}_i)) \\ &= \sum_{i=1}^m l(y_i, \tilde{f}(\vec{x}_i)) \end{aligned}$$

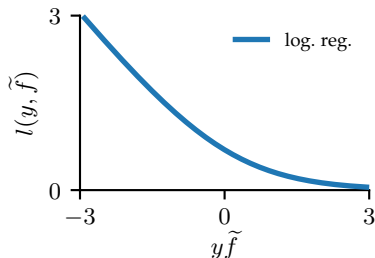
$$\underbrace{l(y, \tilde{f})}_{\text{per-item loss}} = -\ln \sigma(y \tilde{f})$$

Binary Classification Loss

Logistic regression loss is

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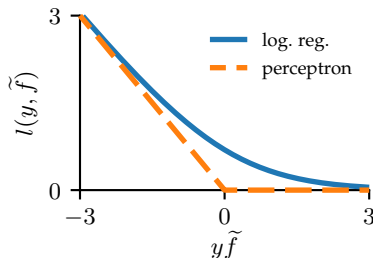


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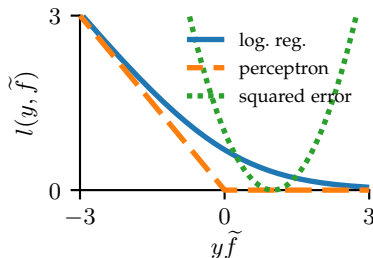


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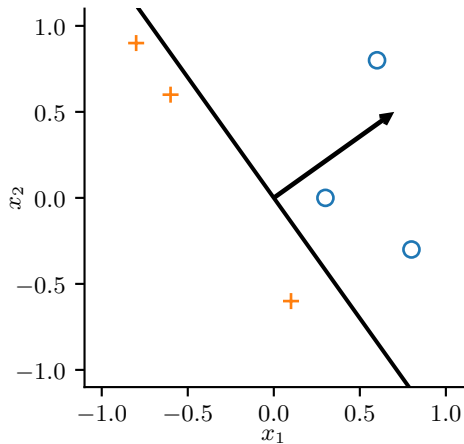
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$$\underbrace{l(y, \tilde{f})}_{\text{per-item loss}} = -\ln \sigma(y \tilde{f})$$



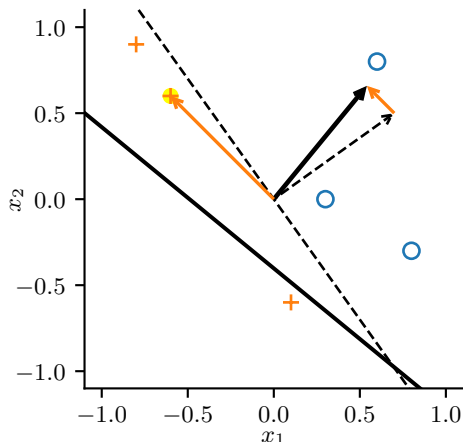
Logistic Regression Example

Stochastic Gradient Descent



Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0 \quad 0.7 \quad 0.5]^T$$

$$y_i = +1$$

$$\vec{x}_i = [1 \quad -0.6 \quad 0.6]^T$$

$$\vec{w}^T \vec{x}_i = -0.12$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.47$$

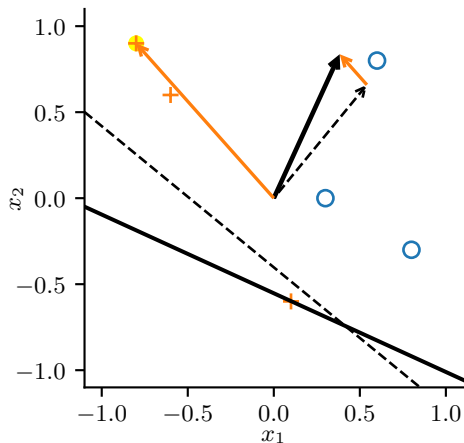
$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.47$$

$$(1 - p_i) y_i \vec{x}_i = [0.53 \quad -0.32 \quad 0.32]^T$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i) y_i \vec{x}_i \\ &= [0.26 \quad 0.54 \quad 0.66]^T \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.26 \quad 0.54 \quad 0.66]^\top$$

$$y_i = +1$$

$$\vec{x}_i = [1 \quad -0.8 \quad 0.9]^\top$$

$$\vec{w}^\top \vec{x}_i = 0.43$$

$$\sigma(\vec{w}^\top \vec{x}_i) = 0.60$$

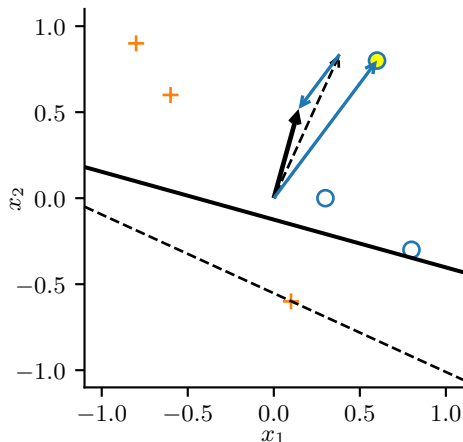
$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i) = 0.60$$

$$(1 - p_i)y_i \vec{x}_i = [0.40 \quad -0.32 \quad 0.36]^\top$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i \\ &= [0.46 \quad 0.38 \quad 0.84]^\top \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.46 \quad 0.38 \quad 0.84]^T$$

$$y_i = -1$$

$$\vec{x}_i = [1 \quad 0.6 \quad 0.8]^T$$

$$\vec{w}^T \vec{x}_i = 1.36$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.80$$

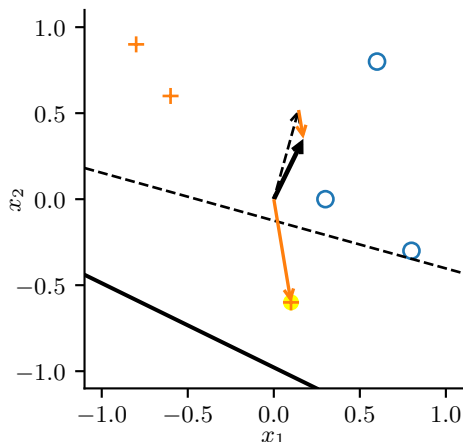
$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.20$$

$$(1 - p_i)y_i \vec{x}_i = [-0.80 \quad -0.48 \quad -0.64]^T$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i \\ &= [0.06 \quad 0.14 \quad 0.52]^T \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.46 \quad 0.38 \quad 0.84]^T$$

$$y_i = +1$$

$$\vec{x}_i = [1 \quad 0.1 \quad -0.6]^T$$

$$\vec{w}^T \vec{x}_i = -0.23$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.44$$

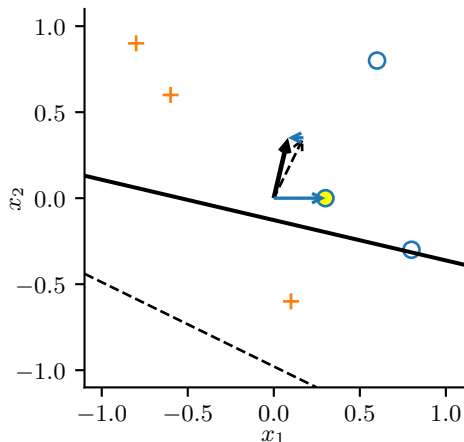
$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.44$$

$$(1 - p_i)y_i \vec{x}_i = [0.55 \quad 0.06 \quad -0.33]^T$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i \\ &= [0.34 \quad 0.17 \quad 0.35]^T \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.34 \quad 0.17 \quad 0.35]^\top$$

$$y_i = -1$$

$$\vec{x}_i = [1 \quad 0.3 \quad 0.0]^\top$$

$$\vec{w}^\top \vec{x}_i = 0.40$$

$$\sigma(\vec{w}^\top \vec{x}_i) = 0.60$$

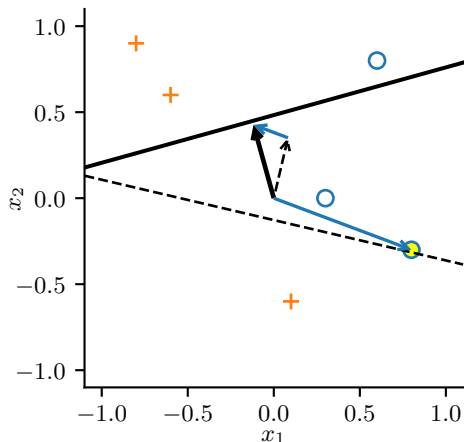
$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i) = 0.40$$

$$(1 - p_i)y_i \vec{x}_i = [-0.60 \quad -0.18 \quad 0.00]^\top$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i \\ &= [0.04 \quad 0.08 \quad 0.35]^\top \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.04 \quad 0.08 \quad 0.35]^T$$

$$y_i = -1$$

$$\vec{x}_i = [1 \quad 0.8 \quad -0.3]^T$$

$$\vec{w}^T \vec{x}_i = 0.01$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.50$$

$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.50$$

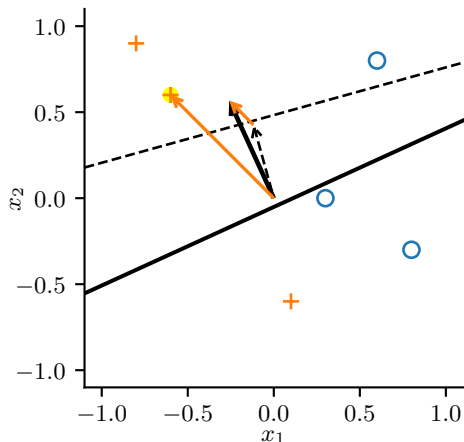
$$(1 - p_i) y_i \vec{x}_i = [-0.50 \quad -0.40 \quad 0.15]^T$$

$$\vec{w} \leftarrow \vec{w} + \eta(1 - p_i) y_i \vec{x}_i$$

$$= [-0.21 \quad -0.12 \quad 0.43]^T$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [-0.21 \quad -0.12 \quad 0.43]^T$$

$$y_i = +1$$

$$\vec{x}_i = [1 \quad -0.6 \quad 0.6]^T$$

$$\vec{w}^T \vec{x}_i = 0.12$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.53$$

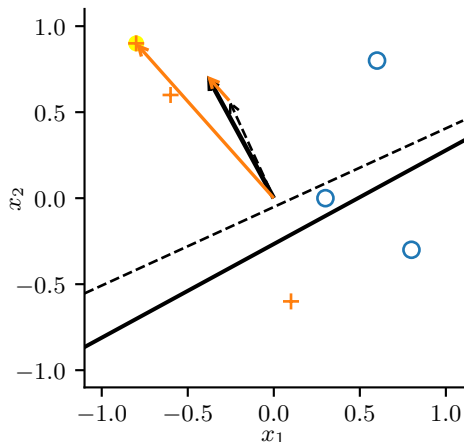
$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.53$$

$$(1 - p_i)y_i \vec{x}_i = [0.47 \quad -0.28 \quad 0.28]^T$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i \\ &= [0.03 \quad -0.26 \quad 0.57]^T \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.03 \quad -0.26 \quad 0.57]^T$$

$$y_i = +1$$

$$\vec{x}_i = [1 \quad -0.8 \quad 0.9]^T$$

$$\vec{w}^T \vec{x}_i = 0.75$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.68$$

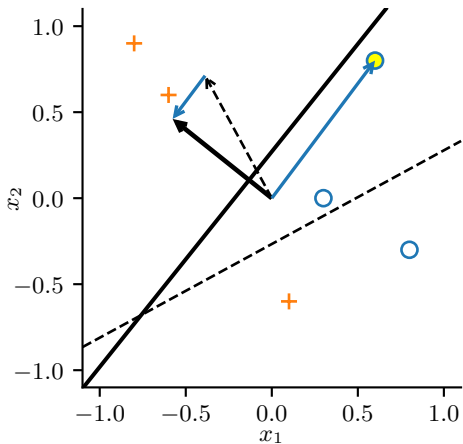
$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.68$$

$$(1 - p_i)y_i \vec{x}_i = [0.32 \quad -0.25 \quad 0.28]^T$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i \\ &= [0.19 \quad -0.39 \quad 0.71]^T \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$\eta = 0.5$

$$\vec{w} = [0.19 \quad -0.39 \quad 0.71]^T$$

$$y_i = -1$$

$$\vec{x}_i = [1 \quad 0.6 \quad 0.8]^T$$

$$\vec{w}^\top \vec{x}_i = 0.53$$

$$\sigma(\vec{w}^\top \vec{x}_i) = 0.63$$

$$p_i = \sigma(y_i \vec{w}^\top \vec{x}_i) = 0.37$$

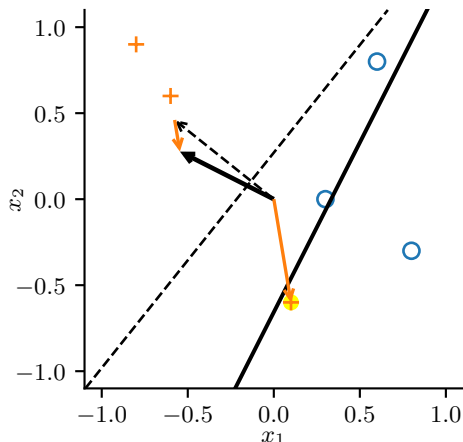
$$(1 - p_i)y_i \vec{x}_i = [-0.63 \quad -0.38 \quad -0.50]^\top$$

$$\vec{w} \leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i$$

$$= [-0.12 \quad -0.58 \quad 0.46]^T$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [-0.12 \quad -0.58 \quad 0.46]^T$$

$$y_i = +1$$

$$\vec{x}_i = [1 \quad 0.1 \quad -0.6]^T$$

$$\vec{w}^T \vec{x}_i = -0.46$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.39$$

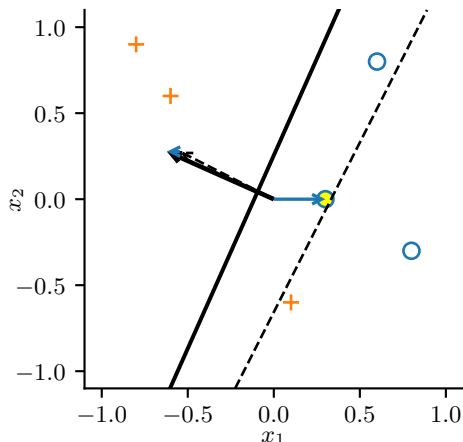
$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.39$$

$$(1 - p_i) y_i \vec{x}_i = [0.61 \quad 0.06 \quad -0.37]^T$$

$$\begin{aligned} \vec{w} &\leftarrow \vec{w} + \eta(1 - p_i) y_i \vec{x}_i \\ &= [0.18 \quad -0.55 \quad 0.28]^T \end{aligned}$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [0.18 \quad -0.55 \quad 0.28]^T$$

$$y_i = -1$$

$$\vec{x}_i = [1 \quad 0.3 \quad 0.0]^T$$

$$\vec{w}^T \vec{x}_i = 0.02$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.50$$

$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.50$$

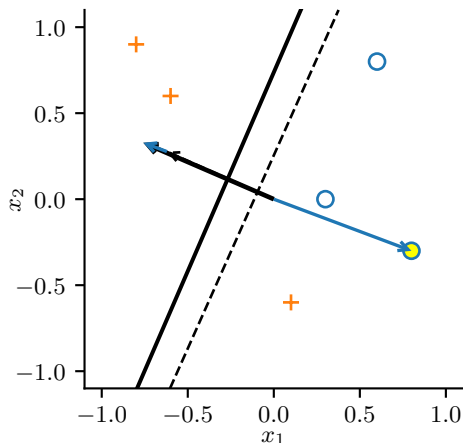
$$(1 - p_i)y_i \vec{x}_i = [-0.50 \quad -0.15 \quad 0.00]^T$$

$$\vec{w} \leftarrow \vec{w} + \eta(1 - p_i)y_i \vec{x}_i$$

$$= [-0.07 \quad -0.62 \quad 0.28]^T$$

Logistic Regression Example

Stochastic Gradient Descent



$$\eta = 0.5$$

$$\vec{w} = [-0.07 \quad -0.62 \quad 0.28]^T$$

$$y_i = -1$$

$$\vec{x}_i = [1 \quad 0.8 \quad -0.3]^T$$

$$\vec{w}^T \vec{x}_i = -0.65$$

$$\sigma(\vec{w}^T \vec{x}_i) = 0.34$$

$$p_i = \sigma(y_i \vec{w}^T \vec{x}_i) = 0.66$$

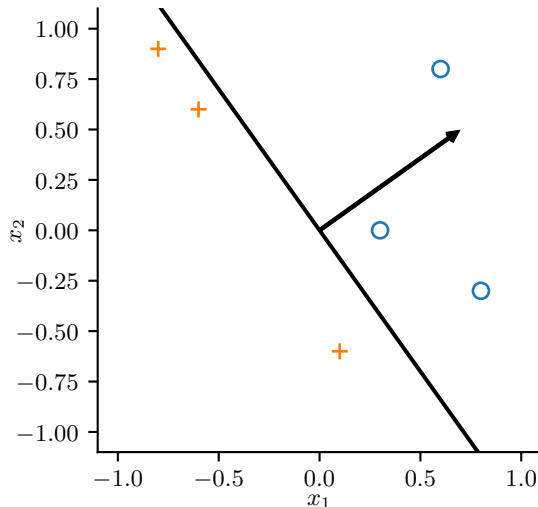
$$(1 - p_i) y_i \vec{x}_i = [-0.34 \quad -0.27 \quad 0.10]^T$$

$$\vec{w} \leftarrow \vec{w} + \eta(1 - p_i) y_i \vec{x}_i$$

$$= [-0.24 \quad -0.76 \quad 0.33]^T$$

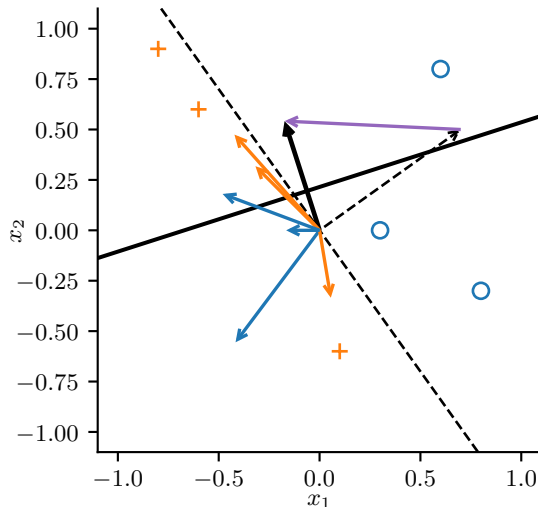
Logistic Regression Example

(Batch) Gradient Descent



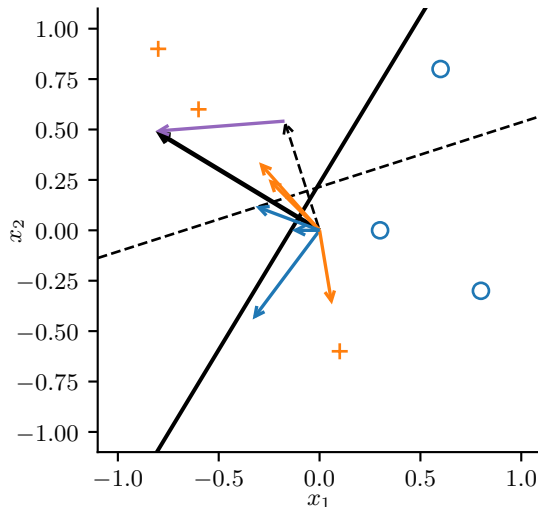
Logistic Regression Example

(Batch) Gradient Descent



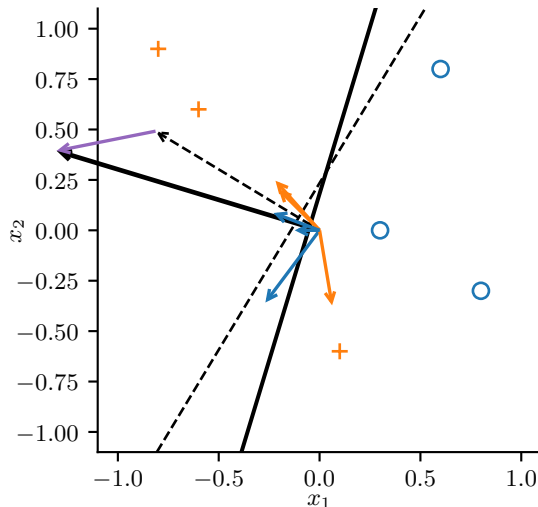
Logistic Regression Example

(Batch) Gradient Descent



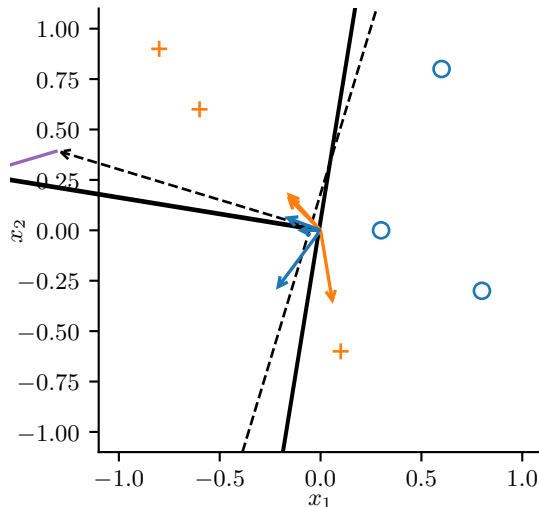
Logistic Regression Example

(Batch) Gradient Descent



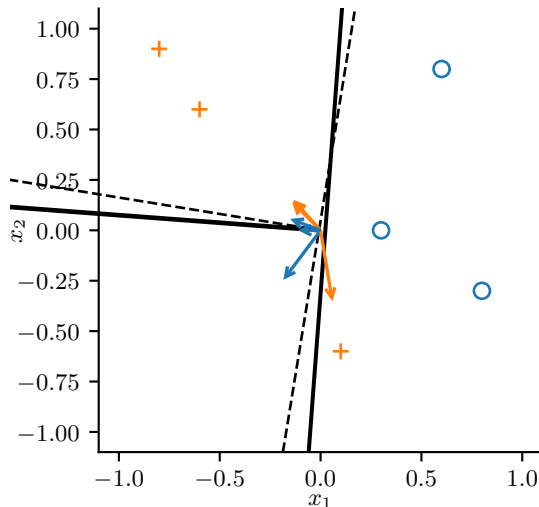
Logistic Regression Example

(Batch) Gradient Descent



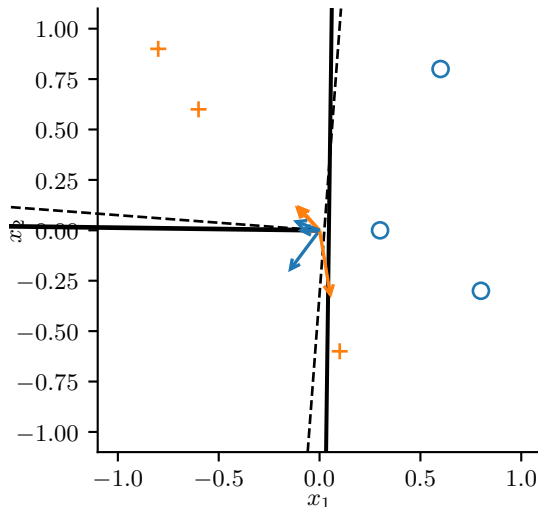
Logistic Regression Example

(Batch) Gradient Descent



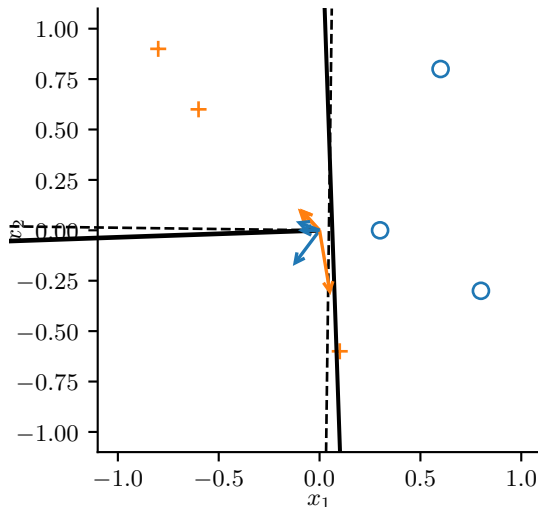
Logistic Regression Example

(Batch) Gradient Descent



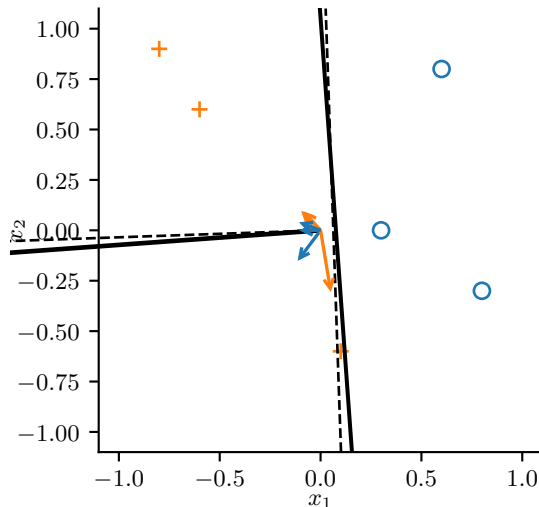
Logistic Regression Example

(Batch) Gradient Descent



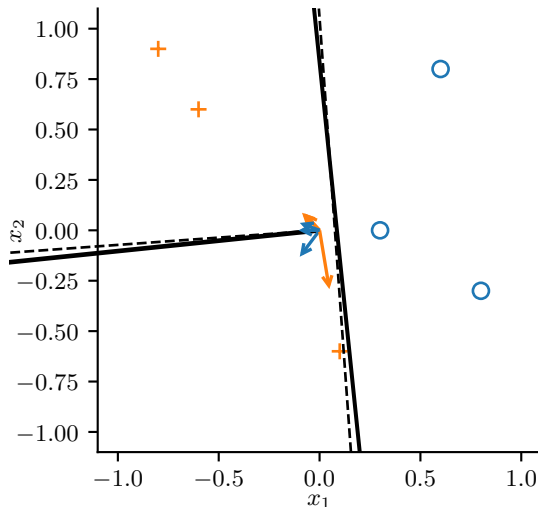
Logistic Regression Example

(Batch) Gradient Descent



Logistic Regression Example

(Batch) Gradient Descent



Logistic Regression Example

(Batch) Gradient Descent

